Time & Work and Pipes & Cesterns

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↳ Time & Work:- The main theme of time & work is to find out "In what time the work is completed under given conditions".

↳ Unit of time is generally based on situation given in the question [It may be sec/min/hr/day/month/yr ...]

↳ Generally in most of the cases day is used.

↳ The main important one to know in this concept is efficiency.

↳ Efficiency:- One unit time work is called efficiency/capacity/strength.

↳ If a person can complete a work in 'n' days then he can do $\frac{1}{n}^{th}$ of work in one day. It is only called efficiency.

Work (1 in general) ⟶ in 'n' days

In 1 day = effs ⟶ $\frac{1}{n}$ work $= \frac{\text{Work}}{\text{Total no. of days}}$

$\Rightarrow$ efficiency $= \frac{1}{\text{Total No. of days.}}$

$\Rightarrow$ Total No. of days $= \frac{1}{\text{efficiency}}$

Inverse related.

↳ If A can complete a task in 'x' days, B can complete the same task in 'y' days. eff of A = A eff = A = $\frac{1}{x}$, B eff = B = $\frac{1}{y}$.

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Combined efficiency $= \frac{1}{x} + \frac{1}{y} = \frac{x+y}{xy}$

Total no of days together $= \frac{1}{\text{Combined efficiency}}$

$$\boxed{\text{Total No of days together} = \frac{xy}{x+y}}$$

When 3 members $\Rightarrow$ T.N.D $= \frac{1}{\frac{xy+y3+3x}{xy3}} = \frac{xy3}{xy+y3+3x}$

etc.

Ex-1: A can do a work in 20 days, B in 40 days then together in how many days?

A $\longrightarrow$ 20 days $\Rightarrow A = \frac{1}{20}$, B $\longrightarrow$ 40 days $\Rightarrow B = \frac{1}{40}$

together eff $= \frac{1}{20} + \frac{1}{40} = \frac{2+1}{40} = \frac{3}{40}$

together T.N.D $= \frac{40}{3} = 13\frac{1}{3}$ days

(or)

$x = 20, y = 40 \Rightarrow$ T.N.D $= \frac{20 \times 40}{20+40}$

$= \frac{40 \times 20}{60_3} = 13\frac{1}{3}$ days.

(or)

Units method: Let's consider total work in units.

Total work in units = LCM (20, 40) = 40 units.

(i) 40 units of work by A in 20 days.

A in one day = $A_{eff} = A = 40/20 = 2$ unit.

(ii) 40 units of work by B in 40 days.

B in one day = $B_{eff} = B = 40/40 = 1$ unit.

Together efficiency = $A_{eff} + B_{eff} = (2+1) = 3$ units.

$\Rightarrow$ together 3 units $\longrightarrow$ 1 days

total 40 units $\longrightarrow$?

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$$\Rightarrow \frac{40\times1}{3} = 13\tfrac{1}{3} \text{ days} = \frac{40}{1+2} = \frac{40}{40/20 + 40/40}$$

(or)

formula from the above understanding:-

$$\text{T.N.D Together} = \frac{\text{LCM (Given individual days)}}{\frac{\text{LCM}}{1^{st}\text{ eff}} + \frac{\text{LCM}}{2^{nd}\text{ eff}} + \cdots\cdots}$$

(or)

↳ The above problem can also be written like...

Ex:2:- A is (20 days $\Rightarrow A = \frac{1}{20}$) twice efficient as B ($A = \frac{1}{20} = 2\times\frac{1}{40}$) and B can complete a task in 40 days. Together in how many days?

Ans- 'A' is more efficient person, 'B' is least eff person.

[When days more $\Rightarrow$ eff less, when days less $\Rightarrow$ eff more]

$$B = \frac{1}{40}, \quad A = 2\times\frac{1}{40} = \frac{1}{20}$$

$$\text{together eff} = \frac{1}{40} + \frac{1}{20} = \frac{1+2}{40} = \frac{3}{40}$$

$$\text{together days} = \frac{1}{\text{together eff}} = \frac{40}{3} = \frac{40}{1+\frac{2}{1}}$$

$$\text{T.N.D together} = \frac{\text{least eff person days}}{1 + \text{No. of times}}$$

(or)

Same problem in different perception:-

↳ 'A' is twice eff B and 'A' can complete a task in 20 days. together no. of days are?

$$\text{together days} = \frac{40}{3} = \frac{2\times20}{1+2}$$

$$\text{T.N.D together} = \frac{\text{No. of times} \times \text{More eff person days}}{1 + \text{No. of times.}}$$